

Isaac Frett

SOFTWARE ENGINEER

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SUMMARY

Self-taught software engineer with 5+ years of experience, usually taking a project from the design through the code, the deploy and the maintenance after it. Python services and data pipelines behind executive reporting at a Fortune 500 retailer, a machine-learning platform for futures research and live execution, and a one-person iOS studio with **three apps on the App Store**. I work across the backend, the data layer, the front end and the release process.

EXPERIENCE

Founder & Lead Software Engineer Jan 2026 – Present

IF Development · Remote

- Run a one-person iOS studio — product design, code, artwork, websites, store listings and releases. Shipped **three apps to the App Store** (Starsplit, Choice Bible, Idle Space Mining) and have kept shipping updates since.
- Wrote custom C++ extensions for the Godot game engine to reach iOS features it does not cover on its own — push notifications, in-app purchases and ads — and set up the whole release process from scratch: code signing, automated builds and App Store review.
- Build and deploy the marketing, privacy and support site for each app as plain static HTML and CSS on GitHub Pages, with the text mirrored from the app repository by a publish script.

Stack Godot · C++ · GDScript · iOS / iPadOS / macOS · StoreKit / IAP · App Store Connect

Software Engineer Jan 2023 – Dec 2025

Lowe's Home Improvement · Charlotte, NC

- Built and maintained the Python web services (Flask, FastAPI) used by **500+ employees**, including the reporting that went up to the executive team.
- Owned the data pipelines that moved and cleaned roughly **150GB of data a month** and fed the React dashboards executives used to make decisions worth **\$15M+**.
- Automated the repetitive parts of our security tooling: threat alerts, patching machines, firewall rules and compliance reports.

Stack Python · Flask · FastAPI · React · SQL · ETL · microservices

Merchandising Service Associate Apr 2022 – Jan 2023

Lowe's Home Improvement · Coralville, IA

- Ran the teams that built in-store product displays. Our electronic price signs only fit one kind of display, so I designed and 3D-printed an adapter on my own time that let the store reuse them on any of them.

PROJECTS

Starsplit — puzzle game for iPhone and iPad 2026 · live

Solo · App Store · starsplit.app

- A match-3 game shipped with **600 levels built by hand across six chapters**, each chapter adding a rule the game had not used before — obstacles, a spreading hazard, a seventh colour, timed levels, and goals that ask players to build pieces rather than clear them.
- Wrote a bot that plays every level thousands of times before release and sets its move limit from the results, so all 600 are tuned from data rather than by hand. Boards are fixed rather than random, so a given level is the same for every player.

Choice Bible — reader for iPhone, iPad and Mac 2026 · live

Solo · App Store · choicebible.com

- **Seven translations and a prebuilt full-text search index ship inside the app**, so search across all 66 books runs locally with no network. Custom text layout makes the page look like a printed book — twelve fonts, text set flush to both edges, automatic word breaks, spacing and margins.
- Notes, highlights, bookmarks and reading position sync across iPhone, iPad and Mac through the user's own iCloud account, which removes the backend entirely — no service to operate, no accounts to manage and no user data held.

Solo · App Store · idlespacemining.app

- Twelve generators with branching upgrade paths, and two levels of reset that trade everything owned for permanent gains.
- Balanced an economy where every tenth generator owned doubles its output, and credited what was mined while the app was closed from the time since it last ran.

Solo · overview: github.com/isaacfrett/quant-public

A machine-learning platform that researched futures trading strategies, validated them, and ran the survivors against a live broker account — the model inputs, the training, testing against past data, execution and monitoring.

- Built roughly **700 model inputs** from 50+ reusable components, with an automated check that none could accidentally see the future — a bug that makes results look excellent and be worth nothing, so finding one blocked the release.
- Strategies were scored only on data they had never seen, and the scoring corrected for how many thousands of ideas had been tried — the difference between a real advantage and luck.
- Trading fees were subtracted before anything was scored, so the bar was "still profitable after costs" rather than "profitable on paper". Two models worked together: one predicted direction, a second decided whether that prediction was worth acting on.
- Simulation and live trading had to behave the same way, checked continuously — any difference stopped a release. Hard limits on how much could be risked per trade, lost in a day and lost overall, plus a kill switch, monitored on a real-time React dashboard.

Stack Python · LightGBM · scikit-learn · Parquet/PyArrow · FastAPI · React + TypeScript · pytest (TDD-first)

Solo · github.com/isaacfrett/statlume-api

- Pulled NBA, NFL and MLB game logs and betting odds from outside providers into a **MySQL warehouse**, refreshed weekly.
- Built models predicting individual player performance — points and rebounds — from how players had been doing lately, and grouped NFL players by how they play rather than ranking them on a single number.
- Generated formatted Excel reports instead of raw tables, with every league built to the same structure.

Stack Python · Django · TensorFlow/Keras · scikit-learn · MySQL · SQLAlchemy · Poetry · pytest

EDUCATION

Bellevue University · graduated, 4.0 GPA

Cybersecurity Club · Phi Theta Kappa Honors Society

SKILLS

LANGUAGES	Python · TypeScript · JavaScript · Java · SQL · C++ · GDScript · Bash
BACKEND	FastAPI · Flask · Django · REST APIs · microservices · ETL pipelines · SQLAlchemy · pytest
FRONTEND	React · TypeScript · Vite · responsive layout · accessibility
MOBILE	iOS · iPadOS · macOS · Godot · StoreKit / in-app purchases · App Store Connect
MACHINE LEARNING	LightGBM · scikit-learn · TensorFlow/Keras · PyTorch · pandas · NumPy · SciPy · PyArrow/Parquet · walk-forward cross-validation with purge/embargo · combinatorial purged CV · meta-labeling · probability calibration · deflated Sharpe · probability of backtest overfitting · feature-leakage auditing
DATA, CLOUD & TOOLS	PostgreSQL · MySQL · AWS · Azure · GCP · Git · CI/CD (Jenkins) · Docker · Kubernetes · Grafana · Power BI